

VERIFICATION OF THE “Mangrove Restoration And Coastal Greenbelt Protection In The East Coast Of Aceh And North Sumatra Province, Indonesia” PROJECT



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Summary:

This report describes the verification audit of the “Mangrove Restoration And Coastal Greenbelt Protection In The East Coast Of Aceh And North Sumatra Province, Indonesia project (“the project”), a reforestation project located in Indonesia against the VCS Version 3 and all applicable requirements therein, for the monitoring period from 14 June, 2011 to 30 September, 2015. The verification was conducted by means of document review, follow-up interviews and site inspection, and the resolution of outstanding issues. The verification included: evaluating whether the provisions of the monitoring methodology and the monitoring plan were consistently and appropriately applied; and assessing the collection of evidence supporting the reported data for conformity to the protocol. As part of the verification 8 findings were raised: 4 Non-Conformity Reports, 4 New Information Requests and no Opportunities for Improvement. These findings are described in Appendix A of this report. In summary, the audit team is able to verify with a reasonable level of assurance that the GHG removals generated by the project, quantified in accordance with the verification criteria, amount to 144,128 tonnes CO₂ equivalent. SCS verified that the non-permanence risk rating of the proposed project activity for this verification is 13%, which is to be applied to the change in carbon stocks at this verification (i.e. equal to 18,737 tCO₂e). The amount of VCUs to be issued would be 125,391 tCO₂e. There are no uncertainties associated with the verification conclusion.

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1 INTRODUCTION

1.1 Objective

The objective of the verification was to conduct an independent assessment of the "Mangrove Restoration And Coastal Greenbelt Protection In The East Coast Of Aceh And North Sumatra Province, Indonesia" project ("the project") to determine whether the quantification of GHG emission reductions or removals for the project during the period 14 June, 2011 to 30 September, 2015 ("the monitoring period") complies with the verification criteria, as set out in the guidance documents listed in Section 1.2 of this report.

1.2 Scope and Criteria

In accordance with Section 4.3.4 of ISO 14064-3:2006, the scope was defined as follows:

- The project and its baseline scenarios;
- The physical infrastructure, activities, technologies and processes of the project;
- The GHG sources, sinks and/or reservoirs that are applicable to the project;
- The types of GHGs that are applicable to the project; and
- The reporting period, as discussed in Section 5 of this report.

In accordance with Section 5.3.1 of the VCS Standard, the criteria for verification was the VCS Version 3, including the following documents:

- AFOLU requirements: VCS Version 3.4;
- AFOLU Non-Permanence Risk tool: VCS Version 3.2; and
- The CDM methodology CDM AR-AM0014 Version 3.0, which is approved for use under the VCS Program.

Unless otherwise indicated, the assessment was performed against the most recent version of the relevant VCS guidance document.

In addition, the assessment was performed against the requirements of the validated project description.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported net anthropogenic GHG removals data is free from material misstatement. SCS applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per VCS Standard Section 5.3.1 4).

1.4 Summary Description of the Project

Project Proponents (Parties):	Livelihoods Fund
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Title of project activity:	Mangrove Restoration And Coastal Greenbelt Protection In The East Coast Of Aceh And North Sumatra Province, Indonesia
Baseline and monitoring methodology	CDM AR-AM0014 Version 3.0
Location of the project activity	The proposed grouped project is located in the East coast of Northern Sumatra island, where it extends from north Banda Aceh sea to the Percut east bank of the Deli river.
Project's crediting period:	14 June, 2015 to 13 June, 2031
Period verified in this verification:	14 June, 2011 to 30 September, 2015

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through a combination of document review, interviews with relevant personnel and on-site inspections, as discussed in Sections 2.2 through 2.4 of this report. At all times, the project was assessed for conformance to the criteria described in Section 1.2 of this report. As discussed in Section 2.5, findings were issued to ensure that the project was in full conformance to all requirements.

Prior to conducting the site visit activities, the audit team created a sampling plan to determine the areas with the greatest risk of material error. The audit team created a sampling plan following a proprietary sampling plan workbook developed by SCS. Per Section 4.4.3 of ISO 14064-3:2006, the audit team identified possible risks of errors, omissions and misrepresentations with respect to the verification criteria. For each identified risk, the audit team assessed the likelihood of the material discrepancy occurring, the likelihood of the material discrepancy not being prevented or detected by the controls of the project the material discrepancy and the likelihood of the material discrepancy not being detected by the audit team. Sampling and data testing activities were planned to address any risk where the likelihood of a material discrepancy not being detected by the audit team was judged to be unacceptably high. The audit team then created a validation plan that took the sampling plan into account.

2.2 Document Review

The monitoring report (version 04 dated 4 December, 2015) was carefully reviewed for conformance to the verification criteria.

The following documents provided by the project proponents have been used as direct sources of evidence for the verification conclusions:

Document name	Ref
- YAGASU & UNIQUE forestry and land use: <i>VCS-PD for project activity</i> “Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province, Indonesia”, version 1.0 dated 21 September 2015 (first version received from the project proponent) and version 3.0 dated 04 December 2015. - YAGASU & UNIQUE forestry and land use: <i>VCS-Monitoring report for project activity</i> “Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province, Indonesia”, version 1.0 dated 30 September 2015 (first version received from the project proponent) and version 4.0 dated 04 December 2015.	/1/
YAGASU & UNIQUE forestry and land use Non-Permanence risk assessment report, version 3.0, 04 December 2015.	/2/
Various entities. Signed contracts and agreements: - Livelihoods Not for Profit Declaration. March 17, 2015 - Purchase Agreement between Yagasu Aceh and Danone. Signed June 30, 2011 - Carbon Emissions Reduction Collaboration agreement between Yagasu Aceh and Livelihoods fund. Signed Sept 18, 2012.	/3/

Document name	Ref
- Community and carbon agreements signed with Community Groups and Village Committees for the following plots: LSKP13P05, TATK11P23, LSSL12P05, TIMA13P141, TIMA13P55, TIPB13P16, TATK12P23, LSSL13P34	
YAGASU & UNIQUE forestry and land use. ESRI Shapefiles - Project boundary including information on location of grouped project regions, planting plot, planting species, and corresponding statistics; - Giri C, Ochieng E, Tieszen LL, Zhu Z, Singh A, Loveland T, Masek J, Duke N (2011). Status and distribution of mangrove forests of the world using earth observation satellite data. Global Ecology and Biogeography 20: 154-159 ; http://data.unep-wcmc.org/datasets/4	/4/
YAGASU and UNIQUE. Evidence of start date: - MoM with villagers and decision on planting campaign start date - Payment transfer receipt to same village for purchasing seedlings and planting	/5/
Ministry of Environment and National Council on Climate Change – Government of Indonesia. Letter of Approval and support for the “Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province, Indonesia”. 29 October 2012	/6/
YAGASU & UNIQUE forestry and land use. GIS data and information: - ESRI Shapefiles with limits of Grouped Project - ESRI Shapefiles with limits of project boundary and land eligibility information - ESRI Shapefiles with location of Sampling Plots	/7/
YAGASU & UNIQUE forestry and land use. Baseline carbon stocks spreadsheet and sampling plot data.	/8/
YAGASU & UNIQUE forestry and land use. Ex-ante GHG removals calculations and ex-ante GHG calculations	/9/
Financial information and data: - Livelihoods Fund: Minutes of the Board of Directors indicating updated business plan. 18 December 2014 - YAGASU Profitability Calculation and cash Flow Spreadsheet: 9 September 2015 - YAGASU Draft Report of Economic Valuation in mangrove Ecosystems in the East Coast of Sumatra. November 2014.	/10/
YAGASU & UNIQUE forestry and land use. Standard Operating Procedures (SOPs): - Pilot Inventory Sampling and Monitoring Plan. 19 April 2015 - Additional SOP and supporting templates: - Planting Mangroves November 2009 - Nursery, Planting, Report for Contributor, Annual Report - Protocol of Carbon Accounting in Restoration Mangroves - Working Paper: CIFOR Protocols for the Measurement, Monitoring, and Reporting of Structure, Biomass, and Carbon Stocks in Mangrove Forests.	/11/
Management and Financial Plan of the project covering 50 years, and stamped by the Provincial Head of North Sumatra Environmental Board to ensure plan has been shared with this entity. August 2015.	/12/
UNIQUE. Report on audit conducted on the plantation areas. 23 July 2015.	/13/
YAGASU & UNIQUE forestry and land use:	/14/

Document name	Ref
- Community Diagnostic and Participatory Rural Appraisal (PRA). November 2014 - YAGASU Draft CCB Report. November 2014 - H.J. RUITENBEEK RESOURCE CONSULTING LIMITED: Due Diligence review of a potential wet carbon project for support under the Danone Fund for Nature (DFN) Wet Carbon Initiative. 6 April, 2011	

The following documents served as supporting documents in order to reach a verification conclusion and they relate to background documents and applicable methodologies, tools and standards:

Name of Document	Ref
CDM Executive Board: <i>Methodology</i> CDM AR-AM0014 "Afforestation and reforestation of degraded mangrove habitats", Version 3.0	/15/
CDM Executive Board: 'Combined tool to identify the baseline scenario and demonstrate additionality in AR CDM project activities' (version 1)	/16/
CDM Executive Board: 'Estimation of non-CO ₂ GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity' (version 04.0.0)	/17/
CDM Executive Board: 'Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities' (version 3.1)	/18/
CDM Executive Board: 'Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (version 4.1.0)	/19/
CDM Executive Board: 'Estimation of the increase in GHG emissions attributable to displacement of pre-project agricultural activities in A/R CDM project activity' (version 2.0.0)	/20/
CDM Executive Board: 'Calculation of the number of sample plots for measurements within A/R CDM project Activities' (version 2.1), Annex 15	/21/
CDM Executive Board: 'Demonstrating appropriateness of allometric equations for estimation of aboveground tree biomass in A/R CDM project activities' (version 01.0.0), Annex 28, EB65	/22/
VCSA: VCS standards: VCS Standard Version 3.5, 25 March 2015	/23/
VCSA: AFOLU Non-Permanence Risk tool: VCS Version 3.2, 4 October 2012	/24/
VCSA: 'Program Definitions: VCS Version 3.5', 8 October 2013	/25/
VCSA: AFOLU requirements: VCS Version 3.4, 8 October 2013	/26/
ISO 14064-3:2006: Greenhouse gases — Part 3: Specification with guidance for the validation and verification of greenhouse gas assertions, First edition, 1 March 2006	/27/
ISO 14065:2007: Greenhouse gases — Requirements for greenhouse gas validation and verification bodies for use in accreditation or other forms of recognitions, First edition, 15 April 2007	/28/
VCSA: <i>Validation and Verification Manual</i> . Version 3.0	/29/
Applicable legislation: - Law No. 5 Year 1960, Land Management - Law No. 9 Year 1985, Fishery - Law No. 5 Year 1990, Conservation of Nature Resource and Ecosystem - Law No. 9 Year 1990, Tourism - Law No. 24 Year 1992, Landuse Planning - Law No. 23 Year 1997, Environmental Management	/30/

• Law No. 22 Year 1998, Autonomy of Local Government • Law No. 41 Year 1999, Forestry • Law No. 27 Year 2007, Management of Coastal Areas and Small Islands • Government Regulation No. 28 Year 1985, Forest Protection • Government Regulation No. 29 Year 1986, Environmental Impact Assessment • Government Regulation No. 20 Year 1990, Water Pollution Control • Government Regulation No. 35 Year 1991, Rivers • Government Regulation No. 45 Year 1992, Autonomy of District Government • Government Regulation No. 31 Year 2004, Fishery • President Decree No. 32 Year 1990, Management of Protected Areas • Regulation of Minister of Forestry No. 37/Menhut-II/2007, Community Forest Management • Decree of Minister of Forestry No. 239/Menhut-V/2007, National Mangrove Task Force 2007	
- IPCC, 2003: <i>Good Practice Guidance for Land Use, Land-Use Change and Forestry</i>, prepared by the National Greenhouse Gas Inventories Programme, Jim Penman, Michael Gytarsky, Taka Hiraishi, Thelma Krug, Dina Kruger, Riitta Pipatti, Leandro Buendia, Kyoko Miwa, Todd Ngara (eds). Published: IGES, Japan. URL: http://www.ipcc-nggip.iges.or.jp/public/gpplulucf/gpplulucf.html - IPCC (2006): <i>2006 IPCC Guidelines for National Greenhouse Gas Inventories, Prepared by the National Greenhouse Gas Inventories Programme</i>. Eggleston H.S., Buendia L., Miwa K., Ngara T. and Tanabe K. (eds). Published: IGES, Japan - Including 2013 Supplement to the 2006 IPCC Guidelines for national Greenhouse Gas Inventories: Wetlands – Methodological guidance on organic and wet soils across IPCC land-use categories.	/31/
Forest Carbon Partnership Facility: http://www.forestcarbonpartnership.org/fcp/	/32/
UN-REDD programme: http://www.un-redd.org/	/33/
Duke et al. 2013. Preliminary assessment of biomass and carbon content of mangroves in Solomon Islands, Vanuatu, Fiji, Tonga and Samoa A Report for the MESCAL Project, IUCN Oceania Office, Suva Report No. 13/24	/34/
Komiya et al. 2008. Allometry, biomass and productivity. Mangrove ecology. Aquatic Botany	/35/
Ministry of Forestry, Republic of Indonesia Jakarta. Mainstreaming Mangroves: Proceedings of Regional Symposium on Mangrove Ecosystem Management in Southeast Asia, Feb. 27 – Mar. 1, 2013 Surabaya, Indonesia	/36/
Analuddin et al. 2015. Aboveground biomass status and management effort of unprotected mangrove forest at the surrounding areas of Rawa Aopa Watumohai National Park, Indonesia Conference paper.	/37/
Jachowski et al. 2013. Mangrove biomass estimation in Southwest Thailand using machine learning. Applied Geography.	/38/
Krisnawati et al. 2012. Allometric Models for Estimating Tree Biomass at Various Forest Ecosystem Types in Indonesia. Research and Development Center for Conservation and Rehabilitation, Forestry Research and Development Agency.	/39/

Komiyama 2005. Common Allometric Equations for Estimating the Tree Weight of Mangroves. Cambridge University Press.	/40/
Ong et al. 2004. Allometry and partitioning of the mangrove, <i>Rhizophora apiculata</i> . Forest and Ecology Management.	/41/
Kaufman et al. 2010 Micronesian Mangrove Forest Structure and Tree Responses to a Severe Typhoon. Society of Wetland Scientists.	/42/
Kauffman and Murdiyarso 2011. Addressing climate change adaptation and mitigation in tropical wetland ecosystems of Indonesia. CIFOR Info Brief.	/43/
Anderson and Bowen 2000. Forest Fire Prevention and Control Project Departemen Kehutanankantor Wilayah Propinsi Sumatera Selatan. Fire Zones and the Threat to the Wetlands of Sumatra, Indonesia. European Union, Ministry of Forestry.	/44/
Republic of Indonesia 2007. National Action Plan Addressing Climate Change.	/45/
Republic of Indonesia 2013. National Action Plan for Climate Change Adaptation Synthesis Report.	/46/
Roland Cochard (2011). The 2004 Tsunami in Aceh and Southern Thailand: Coastal Ecosystem Services, Damages and Resilience, The Tsunami Threat - Research and Technology, Nils-Axel Mårner (Ed.), ISBN: 978-953-307-552-5, InTech, Available from: http://www.intechopen.com/books/the-tsunami-threat-research-and-technology/the-2004-tsunami-in-aceh-a-southern-thailand-coastal-ecosystem-services-damages-and-resilience	/47/
Additional Evidence provided for in additionality, baseline, and common practice determination: - Stevenson et al. 1999. Disused Shrimp Ponds and Mangrove Rehabilitation. An International Perspective on Wetland Rehabilitation (p.277-297). Kluwer Academic Publishers. The Netherlands. - Inception Project Report and Study. Yagasu. January 2012. - Field 1998. Rehabilitation of Mangrove Ecosystems: An Overview. Marine Pollution Bulletin Vol. 37, Nos 8-12, pp. 383-392. Elsevier Science Ltd. London, England. - Mangrove reforestation and community development project, North Sumatra, Indonesia: Lessons learned' states that 500 hectares have been restored with 2,5 millions of mangroves propagules; see online: http://www.forestday.org/fileadmin/tropical-workshop/Plenary-2/14A_GarnierP_Mangroves%20reforestation.pdf	/48/
Sukardjo, S. 1992. Biomass and productivity of a <i>Rhizophora mucronata</i> Lamark plantation in Tritih, central Java, Indonesia. Forest Ecology and Management Journal. Vol. 49 No. 3-4 pp. 195-209.	/49/
Amira, S. 2008. Pendugaan biomassa jenis <i>Rhizophora apiculata</i> Bl. di hutan mangrove, Batu Ampar, Kabupaten Kubu Raya, Kalimantan Barat. Skripsi Departemen Konservasi dan Sumberdaya Hutan, Fakultas Kehutanan, Institut Pertanian Bogor, Bogor. Translation: Amira, S. 2008 An estimation of <i>Rhizophora apiculata</i> Bl. biomass in mangrove forest in Batu Ampar Kubu Raya Regency, West Kalimantan. Undergraduate Thesis, Bogor Agricultural University, Indonesia.	/50/
Murdiyaso et al. 2015. The Potential of Indonesian Mangrove Forests for Global Climate Change Mitigation. Nature magazine, published online: 27 July 2015.	/51/

Saenger, P.(2002) Mangrove ecology, silviculture and conservation. Kluwer Academic Publishers	/52/
Zanne et al (2009) Global wood density database. http://datadryad.org/resource/doi:10.5061/dryad.234/1	/53/

2.3 Interviews

Interviews constituted an important component of the audit process. Interviews were conducted in two phases: an opening and scoping conference call held on 6 October, 2015; and a set of interviews conducted during the site visit from 14 October 2015 to 18 October, 2015. The audit team conducted various interviews with the project proponent's staff, staff of other project entities involved in the project, and other stakeholders, such as the Director of the Environmental Board of North Sumatra Province, various leaders from participating community groups and villages committee members from 4 different villages. The list of interviewed persons is detailed as follows:

Name and Role	Affiliation	Date interviewed
Representative Leaders of Community Groups (local government)	Village/Community Groups: Matang Seping, Alur Nunang, Sungai Lueng, Kapa	15 and 16 October, 2015
Mr Bambang Suprayogi (CEO)	Yagasu	14-18 October, 2015
Dr Hidayati (Director, Environmental Board of North Sumatra Province), her staff, and local University Students	Environmental Board of North Sumatra Province	17 October 2015
Meilinda Suriani (Community Development Manager)	Yagasu	14-18 October, 2015
Grace Tanti (Research and Filed Monitoring)	Yagasu	14-18 October, 2015
Rangga Bayu Basuki (Moangrove Restoration and Protection)	Yagasu	14-18 October, 2015
Linggom Sari (Operational Director)	Yagasu	14-18 October, 2015
Jean-Pierre Renaud (General Delegate, Livelihoods Venture)	Livelihoods Fund	6 October, 2015
Stéphane Perrier (Project Manager, Livelihoods Venture)	Livelihoods Fund	6 October, 2015
Matthias Seebauer (Consultant)	UNIQUE Forest and Land Use	6 October, 2015
Prashant Kadgi (Consultant)	UNIQUE Forest and Land Use	14-18 October, 2015

Name and Role	Affiliation	Date interviewed
> 40 members of local communities participating in the project	-	15 and 16 October, 2015

2.4 Site Inspections

The objectives of the on-site inspections performed were to:

- Select samples of data from on-the-ground measurements for verification in order to meet a reasonable level of assurance and to meet the materiality requirements of the project, as required by Section 5.1.3 of the VCS Standard;
- Perform a risk-based review of the project area and project activities to ensure that the project conformed to the requirements of the VCS rules and the methodology throughout the monitoring period; and
- Ensure that monitoring was conducted in accordance with the requirements of the validated monitoring plan, the methodology employed and the VCS rules.

In fulfilment of the above objectives, the audit team performed an on-site inspection of the project area on the dates 14 October, 2015 to 18 October, 2015. The main activities undertaken by the audit team were as follows:

- An assessment of the implementation and operation of the project activity through visual inspection and through interviews with the project proponent's staff. Project boundaries and the stand information were assessed using a Pocket PC with the geographic information uploaded and connected to a GPS receiver;
- Revisiting six randomly selected inventory sampling plots and 15 planting parcels, some of which which were re-measured by the project proponent's staff under observation of the audit team. While the project proponent was carrying out the re-measurement, the audit team verified that the operational and data collection procedures were implemented in accordance with the monitoring plan of the PD and verified the information flows for generating, aggregating and reporting the monitoring parameters. Furthermore, the monitoring equipment was checked in order to confirm that the monitoring practices followed the requirements of the PD and the applicable methodology. Furthermore, the audit team performed a consistency check in order to verify the consistency of the previous measurement and the re-measurement, and to verify the correctness of the reported stand growth.
- Confirmation that the quality control and quality assurance procedures were in place; and
- Interviewed project personnel to confirm the appropriateness of the non-permanence risk score claimed for the project.

2.5 Resolution of Findings

Any potential or actual material discrepancies identified during the assessment process were resolved through the issuance of findings. The types of findings issued by SCS were characterized as follows:

Non-Conformity Report (NCR): An NCR signified a material discrepancy with respect to a specific requirement. This type of finding could only be closed upon receipt by SCS of evidence indicating that the identified discrepancy had been corrected. Resolution of all open NCRs was a prerequisite for issuance of a verification statement.

New Information Request (NIR): An NIR signified a need for supplementary information in order to determine whether a material discrepancy existed with respect to a specific requirement. Receipt of an NIR did not necessarily indicate that the project was not in compliance with a specific requirement. However, resolution of all open NIRs was a prerequisite for issuance of a verification statement.

Opportunity for Improvement (OFI): Receiving an OFI indicates an area that should be monitored or ideally, improved upon. OFI's are often an indication of something that may become a non-conformity if not given proper attention. A root cause analysis and corrective action plan are not required, but highly recommended. OFI's are considered by the audit team to be closed upon issuance, and a response to this type of finding is not necessary.

As part of the verification 4 NCRs and 4 NIRs were issued. All findings issued by the audit team during the verification process have been closed. In accordance with Section 5.3.6 of the VCS Standard, all findings issued during the verification process, and the inputs for their closure, are described in Appendix A of this report.

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period. No Forward Action Requests were issued by SCS.

2.6 Eligibility for Validation Activities

SCS is accredited for conducting validation of AFOLU projects, so this section is not applicable.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

This is not applicable.

3.2 Methodology Deviations

No methodology deviations have been identified.

3.3 Project Description Deviations

No project description deviations were identified.

3.4 Grouped Project

No additional instances were included as part of validation.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

4.1.1 *Implementation status of the project activity(s)*

In order to verify the implementation status and the project boundaries reported in the MR, the audit team conducted an on-site inspection and multiple interviews as described in Sections 2.3 and 2.4 of this verification report. The audit team confirmed that the implementation is in accordance with that stated in the PD, and that no project description deviations were present.

Hence, the audit team is able to confirm that the project implementation is in accordance with the project description contained in the latest version of the PD, dated 04 December, 2015.

4.1.2 *Implementation status of the monitoring plan and the completeness of monitoring*

The audit team checked the monitoring plan contained in the latest version of the PD of 04 December, 2015 and compared it with the MR Version 04 of 04 December, 2015, to verify whether there was any difference that would cause an increase in estimates of the GHG emission reductions in the current monitoring period. The audit team confirmed that the implementation is in accordance with that stated in the monitoring plan of the PD, and that no project description deviations were present.

Monitoring of project implementation

The audit team confirmed that the monitoring of project implementation has been carried out in accordance with the monitoring plan contained in the registered PD. As confirmed by the audit team, forest establishment and management activities are monitored following the specific relevant Standard Operating Procedures (SOPs). These SOPs provide detailed instruction for all activities, indicating clearly how to record the activity and how to report it to the project coordinator. As shown during the site visit, the project proponent effectively monitors all forest establishment activities. Therefore, the monitoring of the forest establishment has been carried out in accordance with the MP contained in the registered PD.

Monitoring of the actual net GHG removals by sinks

The monitoring has been carried out in accordance with the revised monitoring plan and formulae contained in the registered PD of 04 December, 2015.

As required by the monitoring plan of the PD and the applicable methodology CDM AR-AM0014 Version 3.0 the project proponent effectively monitors the required parameters to determine the project's removals by sinks and emissions by sources. The parameters reported, including source, frequency and review criteria as indicated in the monitoring plan were verified to be correct and in line with the registered MP of the PD. Necessary management system procedures including responsibility and authority of monitoring activities have been verified to be consistent with the PD. Knowledge of personnel associated with the project activity was also found to be satisfactory.

4.1.3 *Inclusion under other programs and existence of other forms of environmental credit*

The proposed project activity is an ARR project activity, and it is located in a non-Annex I country under the UNFCCC and a non-Annex A country under the Kyoto Protocol. Therefore, the GHG removals generated would not be part of an emission-trading Program, nor is the project located in a jurisdiction or sector with binding limits.

The proposed project activity is not participating towards any other GHG program as confirmed by the audit team by checking with national authorities as well as checking the CDM, VCS, Gold Standard, and other relevant databases, both at the national level in Indonesia as well as Internationally.

Therefore, the project does not participate under other GHG programs.

The proposed project activity does not generate another form of environmental credit.

The proposed project activity has not been rejected by any other GHG program as confirmed by the audit team by checking the CDM, VCS, Gold Standard, and other relevant databases.

4.1.4 Previously validated methodology deviations

No methodology deviations were identified.

4.2 Accuracy of GHG Emission Reduction and Removal Calculations

4.2.1 Baseline emissions and removals

According to Section 3.1 of the PD, baseline net GHG removals are assumed to be zero as the existing vegetation is not removed from the site, it will not be measured as part of the monitoring and its continuous presence will be monitored too. Hence, the baseline removals = **0 tCO₂**.

4.2.2 Project emissions and removals

The project emissions and removals or, as called in CDM methodologies, actual net GHG removals by sinks ($\Delta C_{ACTUAL,t}$) are equal to the project sum of changes in carbon stocks of the above-ground and below-ground tree biomass ($\Delta C_{P,t}$), minus the sum of the increases in GHG emissions by sources within the project boundary as a result of the implementation of the project activity ($GHG_{E,t}$):

$$\Delta C_{ACTUAL,t} = \Delta C_{P,t} - GHG_{E,t}$$

Changes in carbon stocks ($\Delta C_{P,t}$)

Following the provisions of the applicable methodology, the project sum of changes in living biomass carbon stocks (ΔC_P) would be equal to:

$$\Delta C_{P,t} = \Delta C_{TREE_PROJ,t} + \Delta C_{SHRUB_PROJ,t} + \Delta C_{DW_PROJ,t} + \Delta SOC_{PROJ,t}$$

Where:

- $\Delta C_{TREE_PROJ,t}$: Change in carbon stock in tree biomass in project in year t; t CO₂-e. This is estimated following the tool for the 'Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (version 4.1.0). Considering equations (12), (13), (14), (16), (12), (13), (14) and (15) of the tool, the total carbon stocks in tree biomass within the project boundary in year t (end of monitoring period), has been estimated through the following equation:

$$\Delta C_{TREE_PROJ,t} = C_{BSL} - \frac{44}{12} \times C_{FTREE} \times A \times \sum_{i=1}^M w_i \times \left(\frac{\sum_l f(x_{1,l}, x_{2,l}, \dots) \times (1 + R_j)}{A_{PLOT,i}} \right)$$

Where:

- CF_{TREE} = Carbon fraction of tree biomass; tC t d.m.⁻¹. The project has applied a value of 0.47 as specified in the PD.
- R_j = Root-shoot ratio for tree species j; dimensionless. The project has applied the equation $R_j = \exp[-1.085 + 0.9256 \cdot \ln(AGB)] / AGB$ which is prescribed by the tool 'Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (version 4.1.0).
- $f(x_{1,l}, x_{2,l}, \dots)$ = Above-ground biomass of the tree returned by the allometric equation for species j relating the measurements of tree l to the above-ground biomass of the tree; t d.m. In accordance with the tool 'Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (version 4.1.0), for ex-post estimations the project has applied two different equations: Ong et al. (2004) /41/ for Rhizophora and Komiyama et al. (2005) /40/ for other trees; for conservative measures saplings were not taken into account for this verification. Wood densities were obtained from the global wood density data base Zanne et al. (2011) /53/. These equations have been validated following the procedures of the tool 'Demonstrating appropriateness of allometric equations for estimation of aboveground tree biomass in A/R CDM project activities' (version 01.0.0). According to this tool equations must have used trees growing in the same edapho-climatic conditions as the project area and the requirement is that they must be based on at least 30 trees and a coefficient of determination of 0.85. The equations show a coefficient of determination above 0.85 and both used more than 30 trees. Equations have been adjusted in Mangrove ecosystems in South-East Asia and North-America that are in general similar to the project area as these are wetland soils located under tropical climates, i.e. same edapho-climatic conditions. Measurement of these variables is done following specific SOPs /11/. In order to verify the data, the audit team required the project staff to re-measure PSPs verifying if the SOPs were correctly implemented and comparing the newly measured values with previous measurements. It was confirmed that no systematic errors were committed, therefore confirming the accuracy of the collected data. No transfer errors exist in this case since field data collectors are used and transfer is automatic so no errors could exist in the transfer. Finally, the audit team checked the GHG benefit spreadsheet and confirmed that the monitored and verified values have been correctly applied to the allometric equations and that the biomass per ha of each PSP have been correctly propagated to all strata and then to all the area.
- A = Sum of areas of the tree biomass estimation strata; ha, which is estimated through A_i . Planted areas are monitoring through specific procedures which provide instructions for the measurement of new planted areas and its verification. The audit team confirmed during the site visit that these procedures were adequately implemented and that the project staff effectively monitors the project area. Furthermore, it was confirmed that sufficient QA/QC procedures are in place, such as internal audits or different levels of quality control (e.g., UNIQUE conducts audits on the delineation of the project area /13/. In order to verify the reported values, the audit team verified the data collected and confirmed that the verified data collected was actually used to estimate the GHG benefits, so no manual transposition errors between data sets were identified. The correctness of the collected information was checked by comparing the geographical information provided in the form of ESRI Shapefiles (SHP) through on-site verification using a GPS receiver. Moreover, Very High Resolution (VHR) imagery was used in order to confirm the accuracy in

the delineation of the project area polygons and the accuracy in the strata delineation. The team confirmed that the strata delineation and project boundary delineation is accurate. Furthermore, consistency checks were conducted between stratum information of the project area SHPs and the stratum information provided by the Permanent Sampling Plots (PSP). Moreover, the procedures for data transfer were confirmed to be sound through confirmation that data used in the calculation of GHG benefits was sourced from the verified SHP.

w_i = Ratio of the area of stratum i to the sum of areas of tree biomass estimation strata (i.e. $w_i = A_i/A$); dimensionless. This is estimated with the value above.

$A_{PLOT,i}$ = Area of sample p in stratum i , ha. The plot area used in the forest inventory was that of a 5m radius as required by the existing SOPs /13/. The audit team confirmed that SOPs were correctly implemented: PSPs are circular and a calibrated cord is used in order to determine the distance of trees to the plot center. In case of borderline trees, a measuring tape is used in order to determine exactly if the tree is in or out. In order to verify the data, the audit team required the project staff to re-measure PSPs and confirmed that this process was conducted accordingly. The audit team confirmed that this plot size has been used in the GHG removal calculation spreadsheet in a consistent manner as well.

C_{BSL} = Baseline carbon stocks, tCO₂. These were sourced from the validated in the PD and are equivalent to 23,669 tCO₂, which includes shrubs and trees.

- $\Delta C_{SHRUB_PROJ,t}$: Change in carbon stock in baseline shrub biomass within the project boundary in year t ; t CO₂-e. This is estimated following the tool for the 'Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities' (version 4.1.0). This has been assumed to be zero for ex-post estimates as all biomass implemented by the project is already considered in the above component. Therefore $\Delta C_{SHRUB_PROJ,t} = 0$.
- $\Delta C_{DW_PROJ,t}$: Change in carbon stock in baseline dead wood biomass within the project boundary in year t ; t CO₂-e. This is estimated following the tool for 'Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities' (version 3.0.0). The project proponent has selected the conservative default-factor set out in Section 6.2 of the tool, which consists in the multiplication of the changes of carbon stocks in tree biomass by a conservative default value. Considering equations (8) and (9):

$$C_{DW_PROJ,t} = C_{TREE_PROJ,t} \times DF_{DW}$$

$$C_{DW_BSL,t} = C_{TREE_BSL} \times DF_{DW}$$

Where:

$C_{TREE_PROJ,t}$ = Carbon stock in trees biomass in stratum i at a point of time in year t , as calculated as explained above; t CO₂e. This has been estimated as explained above.

C_{TREE_BSL} = Carbon stock in trees biomass in stratum i at a point of time in year t , as calculated in the validated PD; t CO₂e. This has been estimated in the validated PD, and is equal to 20,804 tCO₂e.

DF_{DW} = Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass; per cent. For ex-post estimates, a value of 6% has been applied which is the appropriate value for tropical regions < 2000 m of elevation with a rainfall > 1600 mm yr⁻¹ according to the tool for 'Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities' (version 3.0.0).

- $\Delta SOC_{PROJ,t}$: Change in carbon stock in the soil organic carbon (SOC) pool within the project boundary, in year t ; t CO₂-e. This is estimated following equation (4) of the applicable methodology:

$$\Delta SOC_{PROJ,t} = \frac{44}{12} \times \sum_{t=1}^t A_{PLANT,t} \times dSOC_t \times 1 \text{ year}$$

Where:

$A_{PLANT,t}$ = Area of sample p in stratum i , ha. The plot area used in the forest inventory was that of a 5m radius as required by the existing SOPs /11/. The audit team confirmed that SOPs were correctly implemented: PSPs are circular and a calibrated cord is used in order to determine the distance of trees to the plot center. In case of borderline trees, a measuring tape is used in order to determine exactly if the tree is in or out. In order to verify the data, the audit team required the project staff to re-measure PSPs and confirmed that this process was conducted accordingly. The audit team confirmed that this plot size has been used in the GHG removal calculation spreadsheet in a consistent manner as well.

$dSOC_t$ = The rate of change in SOC stocks within the project boundary, in year t ; t C ha⁻¹ yr⁻¹. This is 3.32 t C ha⁻¹ yr⁻¹ as described in the PD.

Considering the above values the change in carbon stocks or carbon removals are equal 144,128 tCO₂e.

Project emissions ($GHG_{E,t}$).

No emissions from fires have been considered in the ex-post calculations as no burning is used to clear the land for planting and no burning has occurred in the monitoring period. This was confirmed by the audit team during the site visit no burning occurs as part of the site preparation within the project boundary and no burning occurs as part of the forest management. Therefore, **$GHG_{E,t}=0$** .

Moreover, considering the above values the project emissions and removals or actual net GHG removals by sinks (ΔC_{ACTUAL}) are equal to **144,128 tCO₂e**.

4.2.3 Leakage

According to the applicable methodology CDM AR-AM0014 Version 3.0 leakage emissions are assumed to occur as a result of increased emissions from displacement of agricultural activities to areas outside the project that lead to deforestation and land use change to agricultural land. The audit team confirmed that these emissions were demonstrated to be de-minimis in the registered PD and were assumed to be zero. Moreover, as assessed in Appendix B of this report, market leakage is equation to zero. Therefore, leakage emissions are equal to **0 tCO₂**.

4.2.4 Net anthropogenic GHG emissions

The audit team is able to confirm that:

- The GHG emission reductions and removals are accurate, including the spreadsheet formulae, conversions and aggregations, and the used data and parameters;
- The methods and formulae set out in the project description for calculating baseline emissions, project emissions and leakage have been followed;
- The default values used in the monitoring report are appropriate.

Hence, the audit team is able to confirm that the GHG emission reductions and removals have been quantified correctly in accordance with the project description and applied methodology and that the net anthropogenic GHG emission reductions in the monitoring period 14 June, 2011 to 30 September, 2015 are equal to 144,128 tCO₂e.

4.3 Quality of Evidence to Determine GHG Emission Reductions and Removals

No significant reporting risks have been identified for the data reported. All the data required for net anthropogenic GHG removals or emissions calculations are obtained following Standard Operating Procedures /11/. The procedures effectively cover: management responsibilities, data monitoring procedures, training procedures, periodic internal audits, management reviews and corrective actions in case of any deviations. As confirmed by the audit team, there are QA/QC procedures in place to check the quality of the data collected, transferred, processed and reported, and to assure that the defined standards are met at all times. Implementation of SOPs and QA/QC procedures was verified by the audit team along with the monitoring data flows from data collection to reporting. These facts provided the audit team with confidence in the high quality of the reported results. In order to reach the required reasonable level of assurance, the audit team conducted spot checks of monitored raw data and conducted general consistency checks crossing different data sources. Spot checks were conducted for every step of the information flow, from data collection up to data processing and reporting. Furthermore, the audit team performed a step by step recalculation of pertinent values. In addition, upon completion of the recalculation activities, the audit team was able to trace each value back to its origin and confirm that no transpositions or miscalculations of the data were present.

Hence, the audit team is able to confirm that sufficient and appropriate evidence is available in order to determine GHG emission reductions and removals.

4.3.1 Non-Permanence Risk Analysis

Following the provisions of paragraph 3.19.2 of the VCS Standard Version 3.5, a non-permanence risk report (NPRR) was prepared following the provisions of the AFOLU Non-Permanence Risk tool: VCS Version 3.2 and submitted to the audit team. The findings and conclusion regarding the non-permanence risk analysis undertaken for the project are summarized below for each risk category and factor. Unless noted otherwise, the audit team agrees with the conclusion stated in the non-permanence risk report.

The findings of the audit team regarding the risk scores applied for each factor are as follows.

Project Management		
Risk Factor	Validation Findings	Risk Rating
a)	The species employed by the project are all either <i>Avicennia</i> or <i>Rhizophora</i> , species that are abundant in the original vegetation in the region, and whose	0

	seedlings are also established in areas that were previously covered with mangrove species. Hence, the audit team agrees that this risk is not relevant.	
b)	As confirmed during the site visit, there is little threat of encroachment from outside actors. The project is located in areas where the uses are compatible with the existence of mangrove forests and in fact mangrove forests enhance these uses (i.e. increase in fish habitat) and local communities are being engaged in different social programs. Hence, the audit team agrees that this risk is not relevant.	0
c)	The audit team confirmed that the project management team includes individuals with significant experience in all skills necessary to successfully undertake all project activities. This was evident during the site visit, when it was confirmed that project areas are managed by a very professional team, which includes senior staff with experience in the management and implementation of the project. Moreover, Livelihoods Fund provides all the necessary technical support in order to ensure the correct management of the existing plantations. Hence, the audit team agrees that this risk is not relevant.	0
d)	As confirmed during the site visit, the project management team has a permanent presence in the project areas and Yagasu has offices, which are located less than a day's travel time from project areas. Moreover, Yagasu has local affiliates and partners in project zones with who they are constantly in communication in case any issue arises. Hence, the audit team agrees that this risk is not relevant.	0
e)	The audit team was able to verify that the Livelihoods Fund has established a strong network among all Livelihoods AFOLU projects to improve overall management, implementation, carbon accounting, evaluation, validation and verification processes. Constant follow-up and support is organized for all projects funded by the proponents and a standardized project carbon monitoring process including monitoring of other benefits (livelihood impacts) has also been set in place. Hence, the audit team agrees that the management team includes individuals with significant experience in AFOLU project design and implementation, carbon accounting and reporting under the VCS Program.	-2
f)	The NPRR discusses the existence of an adaptative management plan and evidence has been provided to support it. The audit team checked the provided Management and Financial plan /12/ and confirmed that it includes instructions to identify and assess potential risks to the project and requires the implementation of a mitigation plan. Moreover, evidence was provided that periodical PRAs were conducted /14/ which are used in order to identify any potential issue. As confirmed by the audit team and stated in such plan, the exact procedures are provided in the MoU between Yagasu (implementing partner) and Livelihoods Fund (project proponent) for the monitoring and mensuration of impacts of the project /3/. This contract includes the responsibilities of each of the parties and includes specific procedures for the	-2

	monitoring of the project (e.g. every three years an assessment by Livelihoods Fund will be conducted in order to confirm the status of the plantation and any mitigation action required), which includes the identification of risks and the implementation of the necessary mitigation actions. As confirmed by the audit team, any potential mitigation action is already budgeted in the contract, and as confirmed during the site visit, this adaptative management plan is implemented. Therefore, the audit team agrees that the project can claim the mitigation score of -2 due to the existence of an adaptative management plan.	
Total Project Management (PM) [as applicable, (a + b + c + d + e + f)]		-4
Total may be less than zero.		

Financial Viability		
Risk Factor	Validation Findings	Risk Rating
a)	The audit team checked the revised spreadsheet with the project financials and confirmed that all assumptions were supported by documentary evidence such as contracts, and that conservative assumptions were applied /10/. Hence, the audit team deems the provided financials to provide a conservative estimate of expected cash flows for future years considering socio-economical local circumstances. According to the financials, the break-even point would be reached in year 2021, which is between 4 and 7 years from the current assessment. Therefore, the audit team agrees that the risk would be 1 for this situation.	1
b)		
c)		
d)		
e)	According to the financial evidence provided, the audit team deems that the Livelihoods Fund should not have any problem in meeting this cash-out amount since it is participated in by large companies (e.g. La Poste, SAP AG, Danone) who have plenty of private equity for investing and are in good financing standing according to the audit team competence. The fund is participated in by large companies who decide in board meetings whether or not to invest in a given project. If an investment is decided upon, each entity will commit to the expenditures for the entire duration of the project pro-rata to their participation in their fund. The final disbursement is done on an annual basis approved by a board meeting. This was confirmed by the audit team through the cash flow analysis done in September 2015, where the total investment of the Fund for the project was estimated to be around 3,9 Mio Euros, and in year 1-3 more than 30% of the investment had already been made. During breakeven, more than 80% is already invested. The fund assumes for the financials that the carbon credits generated have a market price, but they use them for compensating for their own emissions as part of their voluntary commitments. Both the funding and the revenues may be	0
f)		
g)		
h)		

	considered secured revenue, which means that the 100% of the cash-out until the project reaches break-even would be covered.	
i)	As assessed above, the project proponent has enough callable financial resources of more than 50% of the total cash-out. The fund is participated by large companies who decide in board meetings whether to invest or not in a project. If an investment is decided, each entity will commit to the expenditures for the entire duration of the project pro-rata to their participation in their fund. The final disbursement is done in an annual basis approved by another board meeting but the funds are already allocated to the project based on the approved investment plan. This was confirmed by the audit team through the cash flow analysis done in September 2015, where the total investment of the Fund for the project was estimated to be around 3,9 Mio Euros, and in year 1-3 more than 30% of the investment had already been made. During breakeven, more than 80% is already invested. The fund assumes for the financials that the carbon credits generated have a market price, but they use them for compensating for their own emissions as part of their voluntary commitments. This commitment can be mobilized annually, so it is secured funding that is readily available to the project, therefore being in compliance with Section 2.2.2 (5) of the AFOLU Non-Permanence Risk Tool. Hence a mitigation factor of -2 may be claimed.	-2
Total Financial Viability (FV) [as applicable, ((a, b, c or d) + (e, f, g or h) + i)]		0
Total may not be less than zero.		

Opportunity Cost		
Risk Factor	Validation Findings	Risk Rating
a)	As assessed in Section Error! Reference source not found. of this validation report, the baseline scenario is the continuation of the land-use prior to project start, i.e. continuation of degraded mangrove with very little activity from local communities (i.e. prawn-seed collection). The only option would be to deforest these newly established forests, which cannot occur as confirmed by members of the Environmental Board of North Sumatra and local communities and government. Hence, there are very few baseline activities, and where they exist, these are subsistence-driven, which indicates an opportunity risk at least of zero. In this respect, the project activity does not represent an alternative land use activity rather than an improvement of the traditional fishery system, which will bring significant increase, and more stable flows of income to those communities. The project will restore the mangrove, which will increase the fish habitat and its population. Hence, the project will generate positive	0
b)		
c)		
d)		
e)		
f)		

	community benefits. This was confirmed through interviews with stakeholders who confirmed this extent. Hence a risk of zero is appropriate.	
g)	Project proponent is not a for-profit organization; they are carrying-out investments in order to use the carbon credits for compensating their own emissions, yet the main driver of their investments are the derived community and development benefits.	-2
h)	The project has in place signed agreements with the different Rural Communities /3/ which state that they commit to conserve the carbon stocks until tree maturity and that they will respect the project activities for the whole project's length. This agreement is a legally binding commitment to continue management practices that protect the credited carbon stocks over the length of the crediting period as required by Section 2.2.3 of the AFOLU Non-Permanence Risk Tool. Moreover, as confirmed by the audit team, once the plantations reach maturity they would be protected by existing laws which do not allow any deforestation of Mangroves /30/. This was effectively confirmed during the interview held with the representative of the Environmental Board of North Sumatra. In view of this, the audit team confirms that the project may claim the mitigation factor.	-2
i)	This has not been argued.	0
Total Opportunity Cost (OC) [as applicable, (a, b, c, d, e or f) + (g+h or i)]		-4
Total may be less than 0 as indicated by the VCSA in the "Errata and clarifications" dated 24 July 2014 (c.f. http://www.v-c-s.org/sites/v-c-s.org/files/Errata%20and%20Clarifications%2C%20AFOLU%20Non-Permanence%20Risk%20Tool%2C%20v3.2_0.pdf)		
Project Longevity		
Risk Factor	Validation Findings	Risk Rating
a)	The project has in place signed agreements with the different Rural Communities /3/ which state clearly that they commit to conserve the carbon stocks until tree maturity and that they will respect the project activities for the whole project's length. This agreement is a legally binding commitment to continue management practices that protect the credited carbon stocks over the length of the project longevity period as required by Section 2.2.3 of the AFOLU Non-Permanence Risk Tool. Moreover, as confirmed by the audit team, once the plantation reach the maturity they would be protected by existing laws which would not allow any new conversion of existing Mangroves /30/.	5
b)		
	The project proponent has argued a project longevity period of 50 years, which is consistent with the management and financial plan which includes procedures and an overall budget for the 50 years has been provided by the project	

	proponent /12/. According to the same, the overall timeframe of this project has an application period of at least 50 years since the starting date of the project, which describes: after the first monitoring campaign, a first period of ten years (2015-2024) is covered by a contractual agreements signed between Livelihoods and YAGASU to support capacity buildings of farmers, surveillance and progressive gain of autonomy; the next 20 years are covered by the same agreement which will be renewed every ten years, at least 2 times. The final period is regarded as the community self-efficiency phase where the additional incomes from activities developed with the communities accrue. Communities have a clear interest in the protection and maintenance of the plantations even beyond 50 years due to the revenue generation. Nevertheless, this period might also be covered by further contractual agreements, based on the carbon credits generated during that period'. Moreover, the YAGASU management and financial planning of the project has already been shared with national government institutions receiving official approval letters, i.e. Ministry of Forestry, Environmental Institution north Sumatra province, as well as the LoA from the National DNA /6//12/. Moreover, it was confirmed during the interviews held with local government and village leaders that the agreements signed with them may be renewed as many times as desired. Therefore, the right of use may be maintained for the entire project longevity as required by Section 2.2.4 (2) of the AFOLU Non-Permanence Risk Tool. Therefore, in accordance to the AFOLU Non-Permanence Risk Tool, Section 2.2.4 it may be confirmed that the project longevity is 50 and that the resulting risk is 5 as there is a legal agreement to continue the management practices.	
Total Project Longevity (PL) May not be less than zero		5

Land Tenure and Resource Access/Impacts		
Risk Factor	Validation Findings	Risk Rating
a)	Ownership and resource access/use rights are not held by same entity(s) for this project.	0
b)	The ownership of project sites is 92% as private lands owned by local members of communities and 8% as communal lands that are managed by village governments. The abandoned private lands usually have arisen after intensive shrimping and fishing practices have collapsed. As a result, communities have also embraced the mangrove restoration effort and have agreed that the rights on the carbon credits generated by this restoration are exclusively allocated to the project proponent.	2

c)	As evidenced during the site visit and in meetings with the Environmental Board of North Sumatra, disputes over access/use of rights over the project areas does not exist.	0
d)	As assessed above, there are no disputes over access/use rights in the project area. Hence, the audit team agrees that this risk is not relevant.	0
e)	The audit team agrees that the potential upstream and sea impacts that could undermine issued credits in the next 10 years are irrelevant or expected to be insignificant was demonstrated through the following: a) Most of the project areas are degraded fish ponds (very few river bank mudflats are also included) with average areas around 2-6 ha. These fish ponds have already been established by local communities which means that the potential upstream or sea impacts have been minimized by the location of the ponds itself from the sea as well as typical tidal management structures such as dykes around the ponds. b) The project has in place as part of its management plan, monitoring instructions to ensure that the project areas are monitored and in the case of any issues that could cause mortality on the plantations or some other matter, the project would implement a set of mitigation actions which are budgeted in the plan /3/. Therefore, any negative impact would be identified and addressed. c) The extremely scattered nature of these project area parcels on the East coast of Northern Sumatra island, stretching along 497 km from Aceh to Percut, further minimize the risk of significant impacts on the plantations and carbon stocks. Moreover, as confirmed by the audit team, the project area is divided in hundreds of polygons that are dispersed in different areas that belong to different hydrological systems. This would mean that carbon stocks are not concentrated in one unique system, so in the case of any change in the upstream and sea impacts it would be highly unlikely that these affects the whole project area or a significant part of the area. In view of this, the audit team agrees that this risk is irrelevant in the case of the project activity and agrees to set this risk to zero. Since the NPRR has been updated with this discussion, this finding has been resolved.	0
f)	As explained in the project longevity risk this mitigation factor may be claimed by the project as the project area is protected by legally binding commitment to continue management practices that protect carbon stocks over the length of the project crediting period	-2
g)	Mitigation not argued by the project proponent.	0
Total Land Tenure (LT) [as applicable, ((a or b) + c + d + e+ f+g)]		0
Total may not be less than zero.		

Community Engagement		
Risk Factor	Validation Findings	Risk Rating
a)	YAGASU conducted wide consultations mobilizing community members via the village governing institutions. These consultations involved all villages in and around the project areas as evidenced by /3/ and during interviews on the site visit.	0
b)	YAGASU conducted wide consultations mobilizing community members via the village governing institutions. These consultations involved all villages in and around the project areas as evidenced by /3/ and during interviews on the site visit.	0
c)	The audit team confirmed that the project is delivering net positive impacts on the social and economic well-being of local communities who derive livelihoods from the project area. This was confirmed through different interviews with stakeholders who confirmed that the project will defend them from the effect of flooding and that it will increase fish availability. The project has not caused any harm to any livelihoods as confirmed by the same stakeholders. In view of this, the audit team is able to confirm that the positive community benefits are obvious and that the project may claim a mitigation factor of -5.	-5
Total Community Engagement (CE) [where applicable, (a+b+c)]		-5
Total may be less than zero.		

Political Risk		
Risk Factor	Validation Findings	Risk Rating
a)	The audit team confirmed the mean governance score for Indonesia for the period 2010 - 2014 across the six governance indicators of World Bank Institute's Worldwide Governance Indicators and found it to be between -0.32 to less than 0.19. Therefore, the audit team agrees that the political risk in this case is 4.	4
b)		
c)		
d)		
e)		
f)	The audit team checked the websites of the Forest Carbon Partnership Facility /32/ and UN-REDD /33/ and confirmed that Indonesia is receiving REDD+ Readiness funding from the World Bank Forest Carbon Partnership Facility, UN-	-2

	REDD or other bilateral or multilateral donors, and is implementing a REDD+ policy framework covering key components such as GHG credit ownership, clear government authority over REDD+ projects, and/or national measurement, reporting and verification systemspreparation.	
Total Political (PC) [as applicable ((a, b, c, d or e) + f)]		2
Total may not be less than zero.		

Natural Risk		
Risk Factor	Validation Findings	Risk Rating
F	As confirmed during the site visit, this risk is not relevant in mangrove ecosystems due to the presence of water. This was further confirmed by the representative of the Environmental Board of North Sumatra during the interview with her.	0
PD	The project proponent has selected a significance of Minor (5% to less than 25% loss of carbon stock) and a likelihood of "Less than every 10 years". The audit team checked various parcels during the site visit and confirmed that some insects sometimes affect young plants, yet this effect seemed to be minor and not very frequent. Furthermore, pests were deemed unlikely to affect significant shares of the project area due to the scattered nature of the project sites as well as procedures in place by the project proponents to react swiftly to any major potential pest and/or disease problem. Plantation monitoring and self audit reports for 2014 and 2015 /13/ that observed only minor pest attacks where leaves were eaten confirms this but where mortality rates were not affected The reduced frequency in insect attack was also confirmed in interviews with local village and community members. Therefore, a frequency of less than 10 years and a significance of "No Loss" or "Insignificant" would be reasonable in this instance. As confirmed during the site visit, these insects only affect certain species so the use of a mix of species in the plantations also seems to be a good strategy in order to mitigate this risk; hence a mitigation factor of 0.5 may be argued.	2.5
W	The project proponent has selected a significance of minor and a likelihood of "Less than every 10 years". As indicated by the evidence provided in the 2007 National Action Plan Addressing Climate Change in Indonesia /45/, its forests face the threat of forest damage and are prone to natural disatsers and extreme weather events, though various studies show that mangrove forests in particular act to mitigate the impacts of cyclones and other extreme weather events, and carbon stocks are frequently not affected. The project proponents have included further evidence in this respect in showing that mangroves can in fact reduce storm surge water levels by slowing the flow of water and reducing surface waves, and, according to Saenger (2008) /52/, less than 1% of trees are	5

	damaged during storms in mangrove plantations with trees 3-5 m tall, and most of these had recovered within 6 months. This means that a frequency of "Less than every 10 years" is reasonable, while a significance of minor is also deemed acceptable at this time.	
G	The main possible geological risk in the Sumatra region is that it is an area that is prone to earthquakes because of its unique geographic location and because it lies at the juncture of two major tectonic plates. The tremendous 9.1 Sumatra-Andaman earthquake of December 26, 2004, was one of the largest earthquakes ever recorded not only in the area, but also in the world. Significant earthquakes also bring along with them the chance for tsunamis. However, the likelihood of colossal earthquakes like that one is quite rare, and the notion that the likelihood of significant events of this nature can conservatively be put at less than every 10 years, which is what the project proponents have put forth in this instance. They have also alluded to Cochard 2004 /47/, as evidence to suggest that the impacts of earthquakes and giant tsunamis are often mitigated by mangrove ecosystems, and that the scattered nature of the project areas to be included as instances further minimizes the risk of significant impacts on plantations and carbons tocks due to these events. Hence a score of 5 is deemed appropriate for this risk category.	5
ON	Not relevant as confirmed by the audit team during the site visit.	0
Total Natural Risk (as applicable, F + PD + W + G + ON)		12.5

In summary, the overall risk rating that was determined for the project, in accordance with the VCS Non-Permanence Risk Tool, is 13%. The audit team has concluded that the above risk rating is in conformance with the VCS rules.

Risk Category	Rating
a) Internal Risk	0
b) External Risk	0
c) Natural Risk	12.5
Overall Risk Rating (a + b + c)	13 %

The audit team confirmed that the non-permanence assessment has been carried out adequately and applying conservative assumptions where needed.

Therefore, the total number of buffer credits are: Buffer credits = 13%* **144,128** = 18,737 tCO₂e in the monitoring period, and the total number of GHG credits eligible to be issued as VCUs for this first project instances of this monitoring period is 125,391 tCO₂e.

5 VERIFICATION CONCLUSION

The project “Mangrove Restoration And Coastal Greenbelt Protection In The East Coast Of Aceh And North Sumatra Province, Indonesia” conforms to the verification criteria for projects and their GHG emission reductions or removals set out in the VCS rules. The audit team asserts no qualifications or limitations with respect to the statement above.

The audit team has been able to confirm that the project and the project monitoring system have been implemented as described in the PD.

The audit team has been able to confirm, with a reasonable level of assurance, that the quantity of GHG emission reductions and removals set out below has been quantified in accordance with the VCS rules. As documented in Appendix B, the audit team can also confirm that the non-permanence risk score of 13% has been quantified in accordance with the VCS rules.

Reporting period: From 14 June, 2011 to 30 September, 2015.

Verified GHG emission reductions or removals in the above reporting period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
14 June, 2011 to 30 September, 2015	0	144,128	0	144,128
Total	0	144,128	0	144,128

The audit team verified that the non-permanence risk rating of the proposed project activity for this verification is 13% which is to be applied to the change in carbon stocks at this verification 144,128 tCO₂, giving a total buffer equal to 18,737 tCO₂. The amount of VCUs to be issued would be **125,391 tCO₂e**.

The number of Buffer credits per vintage year are as follows:

Year	Buffer credits in year t; (tCO ₂ e)
2011	157
2012	1,058
2013	3,547
2014	6,360
2015	7,614

Total	18,737
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APPENDIX A: VERIFICATION FINDINGS

The following tables include all findings issued during the verification audit. It should be noted that all language under “Client Response” is a verbatim transcription of responses provided by project personnel.

NCR 14-06-2011 to 30-09-2015.1 dated 12/9/2015

Standard Reference: AFOLU Non-Permanence Risk Tool, Version 3.2, Section 2.2.3;

Document Reference: 2015-09-18 VCS Non-Permanence Risk Report YAGASU

Finding: a) According to the AFOLU Non-Permanence Risk Tool, Section 2.2.4, Project longevity is the number of years beginning from the project start date that project activities will be maintained, which may be longer than the project crediting period where projects can demonstrate that activities that maintain carbon stocks on which GHG credits have previously been issued will continue beyond the project crediting period. The project proponent has specified a project longevity of 50 years, which in the case of the existence of a legal agreement to continue management practice, would be equal to a risk factor of 5. However, the NPR Report shows a value of zero which is not correct. Please, ensure that the risk factor value is consistent with the explanation provided in the textbox;

Client Response: Numeric values in NPR Report were updated. Risk factor for this part of the internal risks is 5.

Auditor Response: Upon revision of the pertinent numeric values in the updated NPR report, the audit team can now conclude that the correct and corresponding value for this part of the internal risk determination (5) has now been included in the report. Furthermore, this internal risk value determination has now been substantiated with the inclusion of new documented evidence in the form of the financial and management plan of the project, which specifies the project longevity covering 50 years, and which now also includes evidence that the plan has now also been shared with the provisional Head of North Sumatra office.

Closing Remarks: The Client's response adequately addresses the finding.

NIR 14-06-2011 to 30-09-2015.2 dated 12/9/2015

Standard Reference: AFOLU Non-Permanence Risk Tool, Version 3.2, Section 2.2.3.

Document Reference: 2015-09-18 VCS Non-Permanence Risk Report_YAGASU

Finding: The AFOLU Non-Permanence Risk Tool, Section 2.2.3, requires that projects must have financial and management plans covering the whole project longevity and that these are made public or submitted to the local government. The audit team has not received a financial and management plan covering the project longevity of 50 years. Moreover, no evidence has been provided that this has been made public or has been submitted to local government has been provided.

Client Response: The financial and management plan has been shared to the local government. Matthias Seebauer will be attending a mangrove workshop in Medan from 23rd to 26th and will specifically follow up on the publication of this plan.

Auditor Response: Upon revision of new documented evidence provided by the project proponents, the audit team can now not only confirm the existence of a project financial and management plan that covers a project longevity of 50 years, but can also substantiate that the mentioned plan has now also been submitted to local authorities through the presence of an official stamp from the provisional Head of North Sumatra office confirming the reception and review of the plan.

Closing Remarks: The Client's response adequately addresses the finding.

NCR 14-06-2011 to 30-09-2015.3 dated 12/9/2015

Standard Reference: AFOLU Non-Permanence Risk Tool, Version 3.2, Section 2.3.3

Document Reference: 2015-09-18 VCS Non-Permanence Risk Report_YAGASU

Finding: According to the AFOLU Non-Permanence risk tool, these scores have to be averaged over the most recent five years of available data. The NPRR has used data up to 2013, which is not the latest year available; scores for 2014 are already available.

Client Response: Political risks are updated with the latest data (2010-2014). Database is also available as supporting documentation.

Auditor Response: Upon review of the updated Non-Permanence risk tool and report provided for by the project proponents, the audit team can now confirm that these scores have been averaged using the most recent five years of available data, which now include scores for 2014, and for which this new database has also been included as supporting evidence.

Closing Remarks: The Client's response adequately addresses the finding.

NCR 14-06-2011 to 30-09-2015.4 dated 12/9/2015

Standard Reference: VCS Non-Permanence Risks Report Template, Version 3.1.

Document Reference: 2015-09-18 VCS Non-Permanence Risk Report_YAGASU

Finding: According to the VCS Non-Permanence Risk Report Template, Section 4.1, the referred section shall include any deductions for the AFOLU pooled buffer account, if applicable, to determine the number of GHG credits eligible to be issued as VCUs. However, the provided NPRR does not include the number of GHG credits eligible to be issued as VCUs.

Therefore, the NPRR is not in compliance with the VCS Non-Permanence Risk Report Template.

Client Response: Section 4.2 of the Non-Permanence Risk Report includes the number of GHG credits accounted for this Monitoring period, as well as the number of buffer credits and the credits eligible to be issued as VCUs

Auditor Response: Upon review of the updated Non-Permanence Risk Report submitted by the project proponents, the audit team can conclude that Section 4.1 now includes the appropriate deductions for the AFOLU pooled buffer account to determine the total number of GHG credits eligible to be issued as VCUs.

Closing Remarks: The Client's response adequately addresses the finding.

NCR 14-06-2011 to 30-09-2015.5 dated 12/9/2015

Standard Reference: VCS Standard, Version 3.4, Section 3.18.1; VCS Monitoring Report Template, Version 3.3, Initial instructions.

Document Reference: 2015-10-04 Yagasu Monitoring Report.docx

Finding: The VCS Standard, Section 3.18.1 requires project proponents to use the VCS Monitoring Report Template and to adhere to all instructional text within the template. According to the initial instructions, all sections must be completed using Arial 10pt, black, regular (non-italic) font. However, the MR includes some text in a different font or highlighted.

Client Response: Formatting was checked and now is in compliance with the instructions of the Template

Auditor Response: The audit team can now confirm that all sections of the updated monitoring report submitted by the project proponents have now been amended and re-formatted correctly so as to adhere to the specifications of section 3.18.1 of the VCS standard.

Closing Remarks: The Client's response adequately addresses the finding.

NIR 14-06-2011 to 30-09-2015.6 dated 12/9/2015

Standard Reference: VCS Standard Version 3.4, Section 3.1.1 and Section 2.4.1

Document Reference: 2015-10-04 YAGASU_ex-post carbon calculations

Finding: The VCS Standard, Section 3.1.1, requires that projects shall be guided by the principles set out in Section 2.4.1 which include the principle of "Accuracy", i.e. "Reduce bias and uncertainties as far as is practical". The audit team checked the GHG accounting spreadsheet and found that the allometric equations sourced from Komiyama et al. (2005) and Ong et al. (2004) were applied in order to estimate the aboveground biomass per tree. The audit team checked both publications and confirmed that the equations were correctly sourced and that in both cases one of the independent variables is the Diameter at Breast Height (DBH). However, it seems that the equations have been applied to measurements of d30cm (diameter at 30 cm) and d10cm (diameter at 10 cm). Please clarify why these equations have been applied to measurements of d30cm and d10cm.

Client Response: The equation selection was done focusing on the importance of keep a coherence between this verifications and further verifications of the plantations. Since Rhizophora species are the most planted in the project, and stilt roots have a very big influence on the biomass of these trees, an accurate equation for these species was searched. Ong it is not only species specific, but also almost site specific, since it was developed in Malaysia, just opposite of the project area. The calculations to obtain the equation used all kind of diameters, starting on 0,4 cm DBH. In addition, Komiyama et al. (2005) is used as a general equation for different species of mangrove, it is suitable for D30 and no stilt roots are accounted, so we can assume calculations of all other mangrove species in the project are appropriate .

In addition, even though we are aware that in this first verification the use of Ong et al for Rizophora using D30 and D10 may overestimate partially the biomass calculation, seedlings were excluded from the calculation. This means the exclusion of 53% of the individual recorded by the inventories, and a reduction of 20% of the carbon accounted for biomass (according to our calculations, seedlings would increase the carbon stocks at least 7,000 tCO₂e). Therefore we consider biomass estimations accurate and conservative.

Auditor Response: Upon review of the project proponent response, the audit team can now determine as to why the equations in question have been applied to measurements of both d30cm (diameter at 30 cm) and d10cm (diameter at 10 cm). By using both DBH increment for 30 years of estimations and the chosen potential and exponential allometric equations, the project proponents have determined an annual increase in yield reaching its maximum at year 30. While this calculation will not necessarily resemble what would in fact occur in reality (yield will probably be highest at an intermediary age rather than at age 30), the biomass estimates obtained at the end of 30 years is deemed reasonable and conservative. Species densities were obtained from (Zanne et al 2009) and the allometric equations employed were obtained from (Ong et al. 2004), giving both conservative values for the expost calculations.

Closing Remarks: The Client's response adequately addresses the finding.

NIR 14-06-2011 to 30-09-2015.7 dated 12/9/2015

Standard Reference: Methodological tool, "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities" Version 04.2, Appendix 1, "Methods of plot biomass measurement".

Document Reference: 2015-10-21 Yagasu Monitoring Report.docx; 2015-10-04 YAGASU_ex-post carbon calculations

Finding: According to the Methodological tool, "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities" Version 04.2, and in specific, Appendix 1, "Methods of plot biomass measurement", the measurement for fixed area plots for the biomass of individual trees is added and the sum is divided by the area of the sample plot to obtain the plot biomass value. When saplings or seedlings are considered, where the number of saplings with diameter below the range of diameter applicable to the allometric equation is high, the mean biomass of the saplings in a sample plot can be estimated as follows: (1) Determine the diameter mid-way between the diameter of the smallest sapling existing and the smallest diameter allowed by the allometric equation. (2) Harvest from outside the plot area a few saplings having diameter close to the mid-way diameter and obtain the mean biomass per sapling; (3) Count all the saplings in the sample plot and multiply this number by the mean sapling biomass to obtain their contribution to the plot biomass.

The project proponents have included saplings in their estimations for project carbon stocks for trees and shrubs, yet evidence is lacking with regards to the proponents having followed the required steps in determining the mean biomass stemming from saplings, per instructions in the appendix of the methodological tool: (1) obtaining the number of saplings with diameter below the (2) range of diameter applicable to each allometric equations. The project proponents also seem to have selected the third choice for estimating biomass from saplings in a plot: Count all the saplings in the sample plot and multiply this number by the mean sapling biomass to obtain their contribution to the plot biomass; yet no information or values are provided for as to the determination of the (3) mean sapling biomass, only information on seedling height is provided for in the ex-post calculations. Please provide more information to demonstrate that the steps and calculations for obtaining mean biomass for saplings is in conformation with the methodology."

Client Response: Seedlings biomass was excluded from calculation in order to have more conservative calculations for this monitoring period.

Auditor Response: Upon review of the updated monitoring report submitted by the project proponents, the audit team can conclude that they have decided to exclude the calculation of seedling biomass in their calculation of their change in carbon stock from tree biomass, thus insuring the conservativeness of their estimations in this respect.

Closing Remarks: The Client's response adequately addresses the finding.

NIR 14-06-2011 to 30-09-2015.8 dated 12/9/2015

Standard Reference: VCS standard Sections 4.1.7 and 4.5.6

Document Reference: 2015-10-21 Yagasu Monitoring Report.docx; 2015-10-04 YAGASU_ex-post carbon calculations

Finding: After publication of a new mangrove ecosystem assessment study, the project proponents have decided to revise and apply a new dSOC value for their project. According to section 4.1.7 of the VCS standard, where the methodology uses third party default factors and/or standards, such default factors and standards shall meet with the requirements for data set out in Section 4.5.6, mutatis mutandis. The project proponents are requested to provide further justification as to how they adhere to specific requirements 3 and 7 found below.

3. Data shall be from a time period that accurately reflects available technologies and/or current practice, and trends, within the sector. Selection of the appropriate temporal range shall be determined based on the guidance provided in the GHG Protocol for Project Accounting, Chapter 7 (WRI-WBCSD).

7. Data shall be appropriate to the methodology's geographic scope and the project activities applicable under it.

For requirement 3, it is clear the project proponents have utilized the 2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories in Wetlands to arrive at a time period for their analysis. However, the guidelines don't prescribe a time period for analysis, but rather provides guidance on how to calculate one, where "it is suggested to apply the EF for soil carbon accumulation as long as the soil remains rewetted and vegetated, until such time as stocks are equivalent to soil carbon stocks in natural/un-drained settings with vegetation or there is a change in management practice." Using this guidance, one would actually reach a period of more than 100 years with the IPCC default values; hence it is requested that the project proponents further substantiate why they have deemed the chosen time period of 40 years of accumulation as appropriate.

Client Response: Appendix III has been reviewed and further information is provided in order to explain with detail why the calculations are considerate appropriated regarding the requirements of VCS standard.

- Requirement 3: All calculations provided reflect a conservative method to justify the dSOC value, where only 50 cm of soil layer was used for calculation instead of 2 meters, This soil layer was chosen in order to account for the fact that the age of the mangroves in the study sites could not be derived..Just like aboveground carbon, soil and dead root carbon pools increase in size with increasing age (Donato et al. 2011, Alongi 2012). However, while living biomass eventually reaches a dynamic equilibrium, waterlogged mangrove soils continuously keep on accumulating carbon (Hutchison et al. 2013) . Numerous studies have indicated that, with the exception of soils in peatlands, SOC tends to become enriched with soil depth, but usually deeper soil depths need a longer turnover than in the parts where roots biomass have a higher influence . Mangrove root systems are very shallow, therefore the selected

soil layer of 50 cm is used to derive the dSOC value for this project under the assumption that mangrove net primary productivity (including high litter fall rates and fine root turnover rates) is extremely high for young trees until ages of 20 years slowing down significantly after that (see Ong 1993, attached as supporting document).

In addition, for conservative reason the mean values were discounted with 3 standard errors in order to represent the lower bound of the 99% confidence interval. Also, the dSCO value was discounted to account for allochthonous carbon buried at the site following a VCS approved methodology.

- Requirement 7: SOC calculations are based on a recent published study by Murdiyarso et al 2015. The study assessed mangrove ecosystem C stocks of 39 mangroves located in eight sites in Indonesia. This areas share the same geographic and biological characteristics. However, the final SOC value used in this PD still substantially lower than the average value of this publication. Therefore, requirement 7 is fulfilled.

Auditor Response: Upon review of the supplemental information provided for by the project proponents, the audit team can now confirm that the calculations and procedures employed to arrive at a new dSOC default value were done appropriately and in accordance to the requirements of the VCS standard. In addition to the requirements met above, the project proponents also took additional measures to ensure that their factor was constituted using a conservative approach. The main study used to substantiate their soil carbon estimates /51/ uses an analysis in mangroves until 2m of soil depth, yet project proponents only considered 50 cm. Furthermore, while the average AGB carbon stock of the study /51/ is 187.3 tC/ha, and would thus would be achieved after year 27 according to the ex-ante growth model of this project, an accumulation period of 50 years was assumed in this case, more in line with the overall timeframe of the project which has an application period of at least 50 years since the starting date as described in the VCS Non-Permanence Risk Report. Furthermore, Furthermore, the dSOC value was discounted further to also account for any possible allochthonous carbon buried at the site. Based on this, the sequestration values that were derived for soils in the project area can now be deemed to have been obtained through transparent and verifiable information.

Closing Remarks: The Client's response adequately addresses the finding.

NCR 14-06-2011 to 30-09-2015.9 dated 12/18/2015

Standard Reference: Non Permanence Risk Report Template 3.1

Document Reference: 2015-12-04 VCS Non-Permanence Risk Report_YAGASU; 2015-12-04 Yagasu Monitoring Report;2015-12-04 Project Description YAGASU

Finding: The report template states that "Where a risk or mitigation is not relevant to the project, please write "Not applicable"." The risk Tool states that "Each project management risk factor set out in Table 1 shall be assessed. Where a risk factor does not apply to the project, the score shall be zero for such factor."

Whereas the nuance between these two sets of guidance is slight, it is the understanding of the audit team that only when a risk score requirement has no chance of being applicable to the project should the score be "Not Applicable" For example, for non-WRC projects the WERC requirements do not apply.

Where a score does not need to be claimed, however, as such is the case throughout the project risk report the score shall be zero, and this is currently not the case within the project documentation.

Client Response: Risk report revised, all 'not applicable' are set to zero in line with the NPR Excel template

Auditor Response: Template has been revised accordingly. NCR is closed.

Closing Remarks: The Client's response adequately addresses the finding.

NCR 14-06-2011 to 30-09-2015.10 dated 12/18/2015

Standard Reference: ar-am-tool-14-v4.2

Document Reference: 2015-12-04 Yagasu_ex-post carbon calculations

Finding: According to the methodology, the sample statistic ($Z(1-\alpha)$), that was calculated in cell N 4 in the Carbon_Calculations tab should be calculated with $N-m$ degrees of freedom, being N the number of samples and m the number of strata, but this procedure does not seem to have been followed.

Client Response: Value in cell N4 revised accordingly

Auditor Response: The audit team has reviewed this value and can now confirm that the sample statistic has been revised and is now obtained according to the methodology.

Closing Remarks: The Client's response adequately addresses the finding.

NCR 14-06-2011 to 30-09-2015.11 dated 12/18/2015

Standard Reference: ar-am-tool-14-v4.2; Yagasu SOP

Document Reference: 2015-12-04 Yagasu_ex-post carbon calculations

Finding: According to the project proponent's own Standard Operating Procedures, and concurrent with the response given to NIR 7 in this list of findings, saplings and/or seedlings are considered to be biomass with a lower height than that of DBH (1.3m). However, in column J of the table of the "Tree" tab, there are numerous heights below that of 1.3m and that are also labelled as trees. Further information is sought as to the misclassification of these as trees rather than as seedlings, and if they were or were not included in the biomass calculations as expressed in the response to NIR 7 of this list finding.

Client Response: In the Excel sheet '2015-11-13 YAGASU_ex-post carbon calculations' of our second submission, a height analysis of 'trees' as well as 'seedlings' was done in order to provide a justification why seedlings should be also accounted for. This calculation was excluded since it was not 100% in line with the Methodology as well as for conservative reasons. However, the height analysis shows that the 1.3 height 'threshold' for trees is NOT the most important indicator: From 2,369 'seedlings' counted during the inventory, more than 50% (1,409 seedlings) are between 1 and 2 meters high. This is similar to 'trees' where out of a total 2,015 trees, 1,357 trees are in the same height group (1-2 m). During the inventory trainings and the pilot inventory, it was agreed that the most important indicator in the field, is not the tree height, rather than the development of stilt roots, which decides if a mangrove plant is considered a tree or seedling. The established stilt root system of a tree is a sign that the plant has fully adapted to the site conditions after plantation with a very low risk of plant failure. The SOPs needs to be and will be revised accordingly (parts of the SOPs and trainings are standardized for all Livelihoods mangrove projects, therefore this threshold was forgotten to be adapted for Yagasu)

Auditor Response: With regards to saplings vs trees, the methodology in use mentions the following: "Where the number of saplings with diameter below the range of diameter applicable to the allometric equation is high, the mean biomass of the saplings in a sample plot can be estimated as follows: (1) Determine the diameter mid-way between the diameter of the smallest sapling existing and the smallest diameter allowed by the allometric equation. (2) Harvest from outside the plot area a few saplings having diameter close to the mid-way diameter and obtain the mean biomass per sapling; (3) Count all the saplings in the sample plot and multiply this number by the mean sapling biomass to obtain their contribution to the plot biomass." The audit team has reviewed the allometric equations in question, and has determined that because they employed a wide range of diameters (starting from 0,4 cm DBH), that considering biomass smaller than 1.3 m in height for a mangrove tree is appropriate in this context, and that the classification identified by the project proponents is also deemed appropriate for the project, as only a negligible percentage of diameters collected for biomass fall below the 0,4 threshold. During subsequent verifications, assurance that the SOPs have been revised accordingly should be verified.

Closing Remarks: The Client's response adequately addresses the finding.

APPENDIX B: MARKET LEAKAGE ASSESSMENT

In accordance with Section 3.6.4 of the AFOLU Requirements, the quantity of market leakage caused by the project was assessed at verification. The information upon which the market leakage assessment was based is contained within the monitoring report. The market leakage assessment findings and conclusion are as follows.

The methodology CDM AR-AM0014 Version 3.0, which is the methodology used for quantification of GHG emission removals (as stated in the monitoring report) does not identify market leakage as a potential source of leakage.

In summary, the total quantity of market leakage emissions is estimated to be 0 tCO₂e over the monitoring period. The audit team has concluded that the reporting of market leakage emissions is in conformance with the VCS rules and the methodology.